

DR. ARIK JOUKHAJIAN

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Los Angeles, CA

EDUCATION

PhD University of California, Riverside Microbiology Department	2025
BSc University of California, Los Angeles Major in Microbiology, Immunology, and Molecular Genetics Minor in Atmospheric and Oceanic Sciences	2016

LAB EXPERIENCE

UCR, Glassman Lab - Conducting field work in the Mojave Desert for microbial ecology assessments - Analyzing and presenting next-generation sequencing data - Training undergraduate students in applied lab techniques	2020-2025
UCR, Gachomo Lab (Rotation) - Testing methods of fungal DNA extraction from plant roots - Amplification of target fungal sequences for analysis	2020
UCR, Stajich Lab (Rotation) - Assessment of lichenized fungal genomes for enzyme activity - Analysis of secondary metabolite profiles from fungi	2020
Fulgent Genetics BSL-2 Lab Technician - Performed RNA extraction for high-throughput COVID-19 testing - Maintained workflow by assisting with all aspects of sample processing	2020
VRL Eurofins Pre-Transplant Screening BSL-2 Lab Technician - Increased sample throughput by streamlining workflow of automated testing - Reviewed and confirmed donor information - Resolved client issues in a timely and professional manner	2017-2019
UCLA, Research Immersion Laboratory for Microbiology Student - Cultured and sequenced bacterial isolates with anti-fungal compound production - Developed research project from budgeting stage to poster presentation	2015-2016

LAB EXPERIENCE

UCLA, Tina Treude Lab **Research Assistant**

2014-2016

- Sample-processing of deep-sea sediment core samples
- Assisted with sample collection aboard the RV Yellowfin in the Santa Monica Bay

USC, Center for Dark Energy Biosphere Investigations **Research Intern**

2014

- Produced bacterial qPCR data for 40+ deep sea sample sites
- Maintained and prepared new enrichments of microbial samples

PUBLICATIONS

Joukhajian A and Glassman SI, “Eastern Joshua Tree Arbuscular Mycorrhizal Fungi Largely Consistent Across Roots, Soils, and Seasons.” *Environmental Microbiology* 27.7 (2025) <https://doi.org/10.1111/1462-2920.70138>

Joukhajian A, “Uncovering Arbuscular Mycorrhizal Fungi and Microbiome Responses to Dryland Fires”. (2025) Dissertation on ProQuest

Joukhajian A, Pulido Barriga MF, Davis MJ, Sweet LC, and Glassman SI, “Mojave Desert microbial communities show high resistance and resilience over three years despite widespread plant mortality following the Dome Fire.” *Fire Ecology*. In review.

Quaal RJ, Enright DJ, **Joukhajian A**, Zhilik BB, and Glassman, SI. “Effects of Long-Term Fire Retardants on Mycorrhizal Colonization and Soil Microbiomes” *Applied Soil Ecology*. In review.

PRESENTATIONS

Invited Speaker, “Fungal Response to Wildfires”, San Diego Mycological Society. 2025.

Dissertation, “Uncovering Arbuscular Mycorrhizal Fungi and Microbiome Responses to Dryland Fires”, University of California, Riverside. 2025.

Talk, “Eastern Joshua tree arbuscular mycorrhizal fungi across seasons, roots, and soil”, University of California, Riverside. 2025

Invited Speaker, “A Dry Heat: Desert-Dwelling Fungi in California”, Los Angeles Mycological Society. 2024.

Talk, “Monitoring the impacts of the Dome Fire on soil microbial communities in the Mojave Desert”, Ecological Society of America. 2024.

Poster Presentation, “Arbuscular mycorrhizal symbionts of Eastern Joshua trees fluctuate across seasons, roots, and soil”, SoCal Microbiome Symposium 10. 2024.

Poster Presentation, “Arbuscular mycorrhizal symbionts of Eastern Joshua trees fluctuate across seasons, roots, and soil”, International Mycological Congress. 2024.

Talk, “Monitoring the impacts of the Dome Fire on soil microbial communities in the Mojave Desert”, Mycological Society of America. 2023.

PRESENTATIONS

Poster Presentation, “Characterization of the Eastern Joshua Tree Arbuscular Mycorrhizal Community”, Yosemite Symbiosis Symposium. 2023.

Online Talk, “Assessing microbial resilience in post-fire soil communities following the Dome Fire”, University of California, Riverside Microbiology Seminar Series. 2023.

Online Talk, “Mushrooms of Armenia and Beyond.” Hosted by Glendale Community College Science Lecture Series. 2022.

Poster Presentation, “Characterization of the Eastern Joshua Tree Arbuscular Mycorrhizal Community”, Mycological Society of America, Ecological Society of America, and Southern California Microbiome Symposium. 2022.

Online Talk, “What is iNaturalist?” “Ինչ է iNaturalist հարթակը” Hosted by Institute of Botany after A.L. Takhtajyan, National Academy of Sciences, Republic of Armenia, 2021.

Poster Presentation, “Plant Growth Promotion and Fungal Antagonism in Legumes,” UCLA MCDB/MIMG Undergraduate Research Poster Symposium, 2016.

Poster Presentation, “Foliar endophytic fungi diverge from forest-scale island biogeography patterns observed in Yosemite ectomycorrhizal fungi,” American Society of Microbiology World Microbe Forum (Online), 2021. Mycological Society of America Annual Meeting (Online), 2021. Ecological Society of America Annual Meeting (Online), 2021.

Poster Presentation, “Quantification of subseafloor iron reducers and sulfur oxidizers from extinct seamounts,” SACNAS (Society for Advancement of Chicanos/Hispanics and Native Americans in Science), 2014.

COMMUNITY OUTREACH

International Congress of Armenian Mycologists

Co-founder of ICAM, a 501c3 non-profit organization to foster international collaboration and support mycology research in Armenia

Reviewer for Joshua Tree Science Newsletter

Reviewing research submissions for local publications

RUSD Science and Engineering Fair

Judge for RUSD student research projects conducted during 2020

TEACHING

Field Mycology, UCR MCBL 128 with Dr Sydney Glassman

Two quarters as a lab assistant for slide preparation, dichotomous key usage and field identification of mushrooms in Southern California

Microbiology Laboratory, UCR MBCL 121L with Dr Jason Rothman

One quarter as a lab instructor for bacterial culturing techniques, quantification, microscopy, colorimetric testing, staining, and pathogen identification through Koch’s postulates

AWARDS

Publication Grant, Environmental Microbiology \$4,000	2025
UCR Degree Completion Fellowship Award \$20,000	2025
UCR Microbiology Award for Excellence in Mycology \$1,000	2024
Klotz Conference Travel Grant \$800	2024
Los Angeles Mycological Society Invited Speaker \$100	2024
SPUN Underground Explorer Grant (Lead Researcher) \$10,500	2023
HEERF Dissertation Year Fellowship (Spring 2023) \$8,000	2023
Ecological Society of America Opportunity Fund \$600	2022
Shipley-Skinner Reserve – Riverside County Endowment \$21,255	2022
Klotz Conference Travel Grant \$1,140	2022
Sonoma County Mycological Association Award \$1,000	2021
UC Riverside GradEdge Jumpstart \$5,000	2020

SCIENTIFIC SOCIETIES

International Congress of Armenian Mycologists	Since 2020
American Society for Microbiology	Since 2020
Mycological Society of America	Since 2020
Ecological Society of America	Since 2021